

TROUBLESHOOTING

SCORM Learning Object Troubleshooting Guide

Quick Diagnostic Steps

When something isn't working:

1. **Check browser console** (F12 → Console tab)
 - Look for red errors
 - Look for “[SCORM]” messages
 2. **Test locally first** (open HTML file in browser)
 - Does it work outside the LMS?
 - If yes: LMS issue. If no: code issue.
 3. **Check SCORM Cloud** (scorm.com/scorm-solved/scorm-cloud-features/scorm-cloud-sandbox/)
 - Upload to free SCORM Cloud testing environment
 - If works there but not your LMS: LMS-specific issue
 4. **Review test checklist** ([testing/test-checklist.md](#))
 - Ensure all tests passed
-

Common Issues and Solutions

Issue 1: Blank Screen After Launch

Symptoms: - Content launches but shows only white/blank screen - No content visible - May show loading indicator forever

Causes and Solutions:

Cause A: Wrong Launch File Path

How to diagnose: - Check browser console for 404 errors - Look for “index.html not found” or similar

Solution:

```
<!-- In imsmanifest.xml, ensure path is correct: -->
<resource identifier="RESOURCE_1" type="webcontent"
  adlcp:scormtype="sco" href="index.html">
  <!-- NOT href="src/index.html" or href="/index.html" -->
</resource>
```

File structure should be:

```
package.zip
├── imsmanifest.xml
├── index.html      ← Launch file at root
├── scorm-wrapper.js
├── styles/
└── images/
```

Cause B: JavaScript Error on Load

How to diagnose: - Open browser console (F12) - Look for red error messages - Check what line is failing

Solution: - Fix the JavaScript error - Common errors: - Missing semicolons - Undefined variables - Typos in function names

Cause C: Missing Meta Viewport Tag

How to diagnose: - Content loads on desktop but not mobile - Mobile shows zoomed-out tiny version

Solution:

```
<!-- Add to <head> section: -->
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Cause D: LMS Blocking Content in iframe

How to diagnose: - Check console for “X-Frame-Options” or “Content-Security-Policy” errors - Works in SCORM Cloud but not your LMS

Solution: - Contact LMS administrator - May need to whitelist your content domain - May need CSP headers adjusted

Issue 2: SCORM Tracking Not Working

Symptoms: - Content works fine - Completion status never updates in LMS - Scores don’t appear in gradebook

Causes and Solutions:

Cause A: SCORM API Not Found

How to diagnose:

// Check console for:

[SCORM] SCORM API not found. **Running in** standalone mode.

Solution: 1. Verify SCORM wrapper is loading: `<script src="scorm-wrapper.js"></script>`

2. Check file path is correct (relative path!)

3. Verify LMS actually supports SCORM:

- Try uploading a known-good SCORM package
- Contact LMS administrator

Cause B: SCORM API Window Hierarchy Issue

How to diagnose: - Console shows: “Maximum API search attempts reached”

Solution: - This is rare - means LMS iframe structure is unusual - Update scorm-wrapper.js maxAttempts value: `javascript var maxAttempts = 1000; //`
Increase from 500

Cause C: LMSCcommit Not Called

How to diagnose: - Data seems to save during session - Data lost after closing - No errors in console

Solution: Ensure you’re calling commit after setting values:

```
SCORM.set('cmi.core.score.raw', 85);  
SCORM.commit(); // ← Must call this!
```

Or use convenience methods that auto-commit:

```
SCORM.setScore(85); // Auto-commits  
SCORM.setComplete(); // Auto-commits
```

Cause D: Wrong Data Model Element Names

How to diagnose: - Console shows errors like “Invalid data model element”

Solution: SCORM 1.2 uses specific element names:

// CORRECT:

```
SCORM.set('cmi.core.lesson_status', 'completed');  
SCORM.set('cmi.core.score.raw', 85);
```

// WRONG:

```
SCORM.set('cmi.completion_status', 'completed'); // SCORM 2004  
SCORM.set('cmi.score', 85); // Missing .core.score.raw
```

Issue 3: Images Not Loading

Symptoms: - Broken image icons - Alt text shows but no image - Works locally, breaks in LMS

Causes and Solutions:

Cause A: Absolute Paths Used

How to diagnose: - Check HTML source: `html <!-- WRONG: --> ` `` ``

Solution: Use relative paths:

```
<!-- CORRECT: -->


```

Cause B: Case Sensitivity

How to diagnose: - Works on Windows, breaks on Linux LMS server - Filename is “Photo.JPG” but code has “photo.jpg”

Solution: - Make filenames and code match exactly - Best practice: use lowercase for all filenames

```
<!-- File: images/Photo.JPG -->
 ❌ Won't work on Linux

<!-- File: images/photo.jpg -->
 ✅ Works everywhere
```

Cause C: File Not in Package

How to diagnose: - Unzip the SCORM package - Check if image file actually exists

Solution: - Add missing file to package - Rebuild ZIP

Cause D: File Size Too Large

How to diagnose: - Image loads slowly or times out - Other images load fine

Solution: - Optimize image (compress, resize) - Target: < 500 KB per image - Tools: TinyPNG, ImageOptim, Photoshop “Save for Web”

Issue 4: Content Works Locally, Fails in LMS

Symptoms: - Opens fine in browser from filesystem - Breaks when uploaded to LMS

Causes and Solutions:

Cause A: Using localhost or External URLs

How to diagnose:

```
<!-- Check your code for: -->
<script src="http://localhost:3000/app.js"></script>
<link href="https://external-site.com/style.css" rel="stylesheet">
```

Solution:

```
<!-- Bundle everything locally: -->
<script src="app.js"></script>
<link href="style.css" rel="stylesheet">

<!-- OR use reliable CDN (if LMS allows): -->
<script src="https://cdn.tailwindcss.com"></script>
```

Cause B: Cross-Origin Resource Sharing (CORS)

How to diagnose: - Console shows: “CORS policy blocked” - External resources (fonts, APIs) not loading

Solution: - Bundle all resources locally - Or use CDNs that allow CORS (most major CDNs do)

Cause C: Mixed Content (HTTP in HTTPS)

How to diagnose: - Console shows: “Mixed content warning” - LMS uses HTTPS but content loads HTTP resources

Solution:

```
<!-- WRONG: -->
<script src="http://example.com/script.js"></script>

<!-- CORRECT: -->
<script src="https://example.com/script.js"></script>

<!-- OR protocol-relative: -->
<script src="//example.com/script.js"></script>
```

Issue 5: Touch/Mobile Not Working

Symptoms: - Drag-and-drop works on desktop - Doesn't work on iPad/tablet - Buttons don't respond to touch

Causes and Solutions:

Cause A: Missing Touch Event Handlers

How to diagnose: - Test on actual mobile device or mobile emulator - Elements respond to mouse but not touch

Solution: Add touch event handlers:

```
element.addEventListener('mousedown', handleStart);
element.addEventListener('touchstart', handleStart); // ← Add this

element.addEventListener('mousemove', handleMove);
element.addEventListener('touchmove', handleMove); // ← Add this

element.addEventListener('mouseup', handleEnd);
element.addEventListener('touchend', handleEnd); // ← Add this
```

Cause B: Touch-Action CSS Property

How to diagnose: - Touch starts drag but browser also scrolls - Unwanted default touch behaviors

Solution:

```
.draggable-item {
  touch-action: none; /* Disable browser touch handling */
}
```

Cause C: Small Touch Targets

How to diagnose: - Buttons hard to tap on mobile - Users miss targets frequently

Solution: Ensure touch targets are at least 44x44 pixels:

```
button, .clickable {
  min-width: 44px;
  min-height: 44px;
  /* Add padding if needed */
}
```

Issue 6: Accessibility Problems

Symptoms: - Screen reader doesn't announce content - Can't navigate with keyboard - Fails WCAG compliance

Causes and Solutions:

Cause A: Missing Alt Text

How to diagnose: - Screen reader says "image" with no description - Accessibility audit tool flags images

Solution:

```
<!-- WRONG: -->


<!-- CORRECT: -->


<!-- Decorative images: -->

```

Cause B: No Keyboard Navigation

How to diagnose: - Press Tab key - Can't reach interactive elements - Elements don't show focus

Solution:

```
<!-- Add tabindex to custom elements: -->
<div class="button" tabindex="0" role="button"
    onclick="handleClick()">
    Click Me
</div>

<!-- Better: use real buttons -->
<button onclick="handleClick()">Click Me</button>
```

Add visible focus states:

```
button:focus, .interactive:focus {
    outline: 3px solid #3b82f6;
    outline-offset: 2px;
}
```

Cause C: Missing ARIA Labels

How to diagnose: - Screen reader can't identify element purpose - Form fields not associated with labels

Solution:

```
<!-- WRONG: -->
<div>Name</div>
<input type="text">

<!-- CORRECT: -->
<label for="name-input">Name</label>
<input type="text" id="name-input">

<!-- OR use aria-label: -->
<input type="text" aria-label="Enter your name">
```

```
<!-- For complex widgets: -->
<div role="region" aria-label="Quiz questions">
  <!-- content -->
</div>
```

Cause D: Color Contrast Too Low

How to diagnose: - Text hard to read - WebAIM contrast checker fails - Tool:
<https://webaim.org/resources/contrastchecker/>

Solution: - Text on background: minimum 4.5:1 ratio - Large text (18pt+): minimum 3:1 ratio - Interactive elements: minimum 3:1 ratio

```
/* WRONG: */
.text {
  color: #999; /* Light gray */
  background: #eee; /* Almost white - low contrast! */
}

/* CORRECT: */
.text {
  color: #333; /* Dark gray */
  background: #fff; /* White - high contrast */
}
```

Issue 7: Slow Loading

Symptoms: - Content takes > 10 seconds to load - Learner sees loading screen for long time - Timeout errors

Causes and Solutions:

Cause A: Large File Sizes

How to diagnose: - Open DevTools → Network tab - Look for large files (> 1 MB) - Check total package size

Solution: - **Images:** Compress with TinyPNG, use WebP format - **Videos:** Use external hosting (YouTube, Vimeo) instead of embedding - **Fonts:** Use system fonts or limited font weights - **JavaScript:** Minify code

Target package sizes: - Simple interaction: < 5 MB - Rich media: < 20 MB - Complex simulation: < 50 MB

Cause B: Too Many HTTP Requests

How to diagnose: - DevTools Network tab shows 50+ requests - Many small files loading

Solution: - Bundle CSS into one file - Bundle JavaScript into one file - Use CSS sprites for small icons - Use inline SVG for icons

Cause C: Blocking Resources

How to diagnose: - Page hangs waiting for external resource - External CDN is slow or unreachable

Solution:

```
<!-- Add async or defer to scripts: -->
<script src="analytics.js" async></script>
<script src="app.js" defer></script>

<!-- Load critical CSS inline: -->
<style>
  /* Critical above-the-fold styles */
</style>

<!-- Load full CSS asynchronously: -->
<link rel="preload" href="styles.css" as="style"
      onload="this.rel='stylesheet'">
```

Issue 8: Browser-Specific Bugs

Symptoms: - Works in Chrome, breaks in Safari - Works in Firefox, breaks in Edge - Works everywhere except IE11

Causes and Solutions:

Cause A: Modern JavaScript Not Supported

How to diagnose: - Check console in problematic browser - Error like “Promise is not defined” or “async not recognized”

Solution: Use polyfills or transpile code:

```
<!-- Add polyfill for older browsers: -->
<script src="https://polyfill.io/v3/polyfill.min.js?features=es6">
</script>
```

Or avoid modern syntax:

```
// MODERN (may not work in older browsers):
const items = [...array];
const result = await fetchData();

// COMPATIBLE:
var items = array.slice();
fetchData().then(function(result) {
  // handle result
});
```

Cause B: CSS Grid/Flexbox Not Supported

How to diagnose: - Layout broken in IE11 - Elements overlapping or misaligned

Solution:

```
/* Provide fallback: */
.container {
    display: block; /* Fallback */
    display: grid; /* Modern browsers */
}

/* OR use Flexbox (better IE11 support): */
.container {
    display: flex;
}
```

Cause C: Safari-Specific Issues

How to diagnose: - Dates not parsing: `new Date('2023-10-28')` returns Invalid Date -
LocalStorage not working in iframe

Solution:

```
// Safari date parsing:
// WRONG:
new Date('2023-10-28'); // Fails in Safari

// CORRECT:
new Date('2023/10/28'); // Works everywhere
// OR:
new Date(2023, 9, 28); // Month is 0-indexed!

// Safari LocalStorage in iframe:
// May be blocked - check first:
try {
    localStorage.setItem('test', 'value');
    // Works
} catch (e) {
    // Blocked - use alternative (SCORM suspend_data)
    SCORM.saveSuspendData({ test: 'value' });
}
```

Issue 9: Manifest/Package Structure Errors

Symptoms: - LMS rejects package on upload - “Invalid package” error - “Manifest error” message

Causes and Solutions:

Cause A: imsmanifest.xml Syntax Error

How to diagnose: - Open imsmanifest.xml in browser - If XML is invalid, browser will show error

Solution: Validate XML structure:

```
<?xml version="1.0" encoding="UTF-8"?>
<manifest identifier="MANIFEST_1" version="1.0"
  xmlns="http://www.imsproject.org/xsd/imscp_rootv1p1p2"
  xmlns:adlcp="http://www.adlnet.org/xsd/adlcp_rootv1p2"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.imsproject.org/xsd/imscp_rootv1p1p2
    imscp_rootv1p1p2.xsd
    http://www.imsglobal.org/xsd/imsmd_rootv1p2p1
    imsmd_rootv1p2p1.xsd
    http://www.adlnet.org/xsd/adlcp_rootv1p2
    adlcp_rootv1p2.xsd">
  <!-- Content here -->
</manifest>
```

Common errors: - Missing closing tags - Unescaped & characters (use &;) - Unescaped < > characters (use < >)

Cause B: Wrong Folder Structure

How to diagnose: - Unzip package - Check if imsmanifest.xml is at root level

Solution:

CORRECT:

```
package.zip
├─ imsmanifest.xml ← Must be at root!
├─ index.html
└─ ...
```

WRONG:

```
package.zip
├─ my-module/
│   └─ imsmanifest.xml ← Too nested!
└─ ...
```

Cause C: Missing Required Metadata

How to diagnose: - LMS import partially works but missing title/description

Solution: Ensure manifest has required elements:

```
<manifest>
  <metadata>
    <schema>ADL SCORM</schema>
    <schemaversion>1.2</schemaversion>
  </metadata>
```

```

<organizations default="ORG_1">
  <organization identifier="ORG_1">
    <title>Module Title Here</title> ← Required!
    <item identifier="ITEM_1" identifierref="RESOURCE_1">
      <title>SCO Title</title> ← Required!
    </item>
  </organization>
</organizations>

<resources>
  <resource identifier="RESOURCE_1" type="webcontent"
    adlcp:scormtype="sco" href="index.html">
    <file href="index.html"/>
    <!-- List all files -->
  </resource>
</resources>
</manifest>

```

Issue 10: Data Not Persisting

Symptoms: - Learner makes progress - Closes and reopens - Progress lost

Causes and Solutions:

Cause A: Not Using suspend_data

How to diagnose: - Check if code saves to suspend_data - Look for SCORM.saveSuspendData() calls

Solution:

```

// Save progress before exit:
window.addEventListener('beforeunload', function() {
  var progressData = {
    currentQuestion: 5,
    answers: ['A', 'B', 'C', 'D', 'A'],
    score: 80,
    completedSections: [1, 2, 3]
  };

  SCORM.saveSuspendData(progressData);
});

// Load progress on init:
window.addEventListener('load', function() {
  var savedData = SCORM.loadSuspendData(true); // true = parse JSON

  if (savedData) {
    // Restore progress

```

```

        currentQuestion = savedData.currentQuestion;
        answers = savedData.answers;
        // etc.
    }
});

```

Cause B: Exceed 4096 Character Limit

How to diagnose: - Console shows: “Suspend data too large” - Data truncated

Solution: Compress data:

```

// Instead of saving verbose data:
{
    question1: "answer",
    question2: "answer",
    question3: "answer"
}

// Use compact format:
{
    q: ["answer1", "answer2", "answer3"]
}

// Or use compression library (LZ-String)

```

SCORM 1.2 limit: 4,096 characters SCORM 2004 limit: 64,000 characters (consider upgrading if needed)

Cause C: Not Calling LMSCommit

Solution:

```

// After saving suspend data:
SCORM.saveSuspendData(progressData);
SCORM.commit(); // ← Must call this!

```

The scorm-wrapper.js in this framework auto-commits for you, but if using custom code, ensure commit is called.

Diagnostic Tools

Browser Developer Tools

Chrome DevTools (F12): - **Console:** View errors and SCORM logs - **Network:** Check file loading and sizes - **Application:** View LocalStorage, SessionStorage - **Sources:** Debug JavaScript

Firefox Developer Tools: - Similar to Chrome - Better CSS grid inspector

SCORM Cloud Sandbox

Free testing environment: - URL: <https://cloud.scorm.com/sc/guest/SignUpForm> - Upload and test packages - View detailed SCORM logs - If works here but not your LMS → LMS issue

Accessibility Checkers

Browser Extensions: - WAVE (WebAIM) - axe DevTools - Lighthouse (built into Chrome)

Run audit: 1. Open DevTools 2. Go to Lighthouse tab 3. Check “Accessibility” 4. Generate report

Validation Tools

HTML Validator: - <https://validator.w3.org/> - Upload or paste HTML - Checks for syntax errors

XML Validator: - Open imsmanifest.xml in browser - Browser will show errors if invalid

SCORM Manifest Validator: - Some LMS platforms provide this - Or use: <https://github.com/adlnet/SCORM-to-xAPI-Wrapper>

Getting Help

Information to Provide

When asking for help, include:

1. **Package information:**
 - Name and version
 - File size
 - How it was created
2. **Environment:**
 - LMS name and version
 - Browser name and version
 - Operating system
3. **Symptoms:**
 - Exact error messages
 - Screenshots
 - Console logs (copy full text)
4. **What you’ve tried:**
 - Troubleshooting steps already taken
 - Changes made
5. **Reproducibility:**
 - Does it happen every time?
 - Does it affect all users or just some?

Support Channels

Internal: - IT Help Desk: [your ticket system] - Learning Technology Team: [your contact] - LMS Administrator: [your contact]

External: - SCORM.com community forums - LMS vendor support - Stack Overflow (tag: scorm)

Prevention Best Practices

Development Checklist

Before deploying to production:

- ☐ Test locally in browser
- ☐ Test in SCORM Cloud
- ☐ Test in LMS staging environment
- ☐ Test in 3+ browsers
- ☐ Test on mobile device
- ☐ Run accessibility audit
- ☐ Review console for errors
- ☐ Check all image paths
- ☐ Verify SCORM tracking
- ☐ Document any workarounds

Code Quality

- Use relative paths (never absolute)
- Validate HTML and XML
- Test with SCORM wrapper debug mode on
- Add error handling
- Include console logging for debugging
- Comment complex code

Documentation

- Maintain changelog
 - Document known issues
 - Keep testing notes
 - Update metadata in manifest
-

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